Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Half Past Paper – Week 24

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
2	Sharing amounts in a given ratio	4	U577
4	Calculating with rates	4	U256
6a	Probabilities of mutually exclusive events	2	U683
6b	Probability	2	U510,U683
8a	Graphs of direct and inverse proportion	1	U238
8b	Graphs of direct and inverse proportion	1	U238
10	Finding unknown sides in similar shapes	4	U578
12	Index laws	2	U235,U662,U985
14	Combining ratios	4	U921
16	Solving simultaneous equations involving quadratics	4	U547
18	Solving geometric problems using vectors	4	U781
20	Tree diagrams for independent events	2	U558
22	Dividing algebraic fractions	3	U824
	Total	37	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

2 Natalie makes potato cakes in a restaurant.

She mixes potato, cheese and onion so that

weight of potato : weight of cheese : weight of onion = 9 : 2 : 1

Natalie needs to make 6000 g of potato cakes.

Cheese costs £2.25 for 175 g.

Work out the cost of the cheese needed to make 6000 g of potato cakes.

£.....

(Total for Question 2 is 4 marks)

4 A water tank is empty.
Anil needs to fill the tank with 2400 litres of water.

Company **A** supplies water at a rate of 8 litres in 1 minute 40 seconds.
Company **B** supplies water at a rate of 2.2 gallons per minute.

1 gallon = 4.54 litres

Company **A** would take more time to fill the tank than Company **B** would take to fill the tank.

How much more time?
Give your answer in minutes correct to the nearest minute.

..... minutes

(Total for Question 4 is 4 marks)

6 In a bag there are only red counters, blue counters, green counters and pink counters.
A counter is going to be taken at random from the bag.

The table shows the probabilities of taking a red counter or a blue counter.

Colour	red	blue	green	pink
Probability	0.05	0.15	.....	.....

The probability of taking a green counter is 0.2 more than the probability of taking a pink counter.

(*a*) Complete the table.

(2)

There are 18 blue counters in the bag.

(*b*) Work out the total number of counters in the bag.

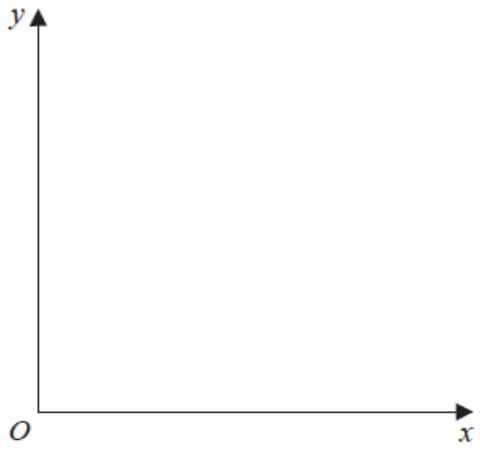
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(2)

(Total for Question 6 is 4 marks)

8 (*a*) Using the axes below, sketch a graph to represent the statement

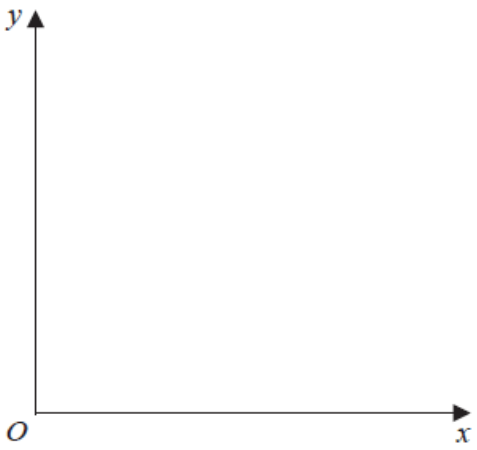
y is directly proportional to *x*



(1)

(*b*) Using the axes below, sketch a graph to represent the statement

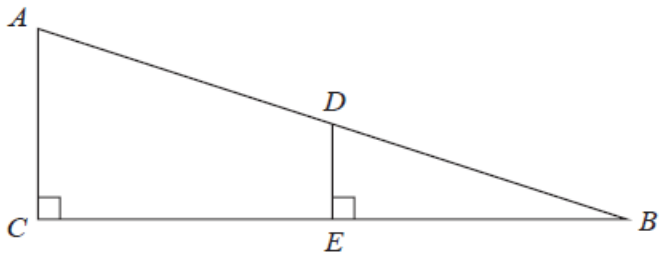
y is inversely proportional to *x*



(1)

(Total for Question 8 is 2 marks)

10 The diagram shows two right-angled triangles *ACB* and *DEB*.



AD = 9 cm
DE = 2 cm
DB = 6 cm

Calculate the length of *CB*.
Give your answer correct to 2 decimal places.

..... cm

(Total for Question 10 is 4 marks)

12 $\left(ax^6\right)^{\frac{1}{n}} = 7x^3$

Work out the value of a and the value of n.

a = ...

n = ...

(Total for Question 12 is 2 marks)

14 Olivia and Jessica have in total half as many sweets as Fran and Gary have in total.

Fran and Gary share their sweets in the ratio 2 : 3
Olivia and Jessica share their sweets in the ratio 9 : 1

Fran got w sweets.
Gary got x sweets.
Olivia got y sweets.
Jessica got z sweets.

Find, in its simplest form, $w : x : y : z$

.....

(Total for Question 14 is 4 marks)

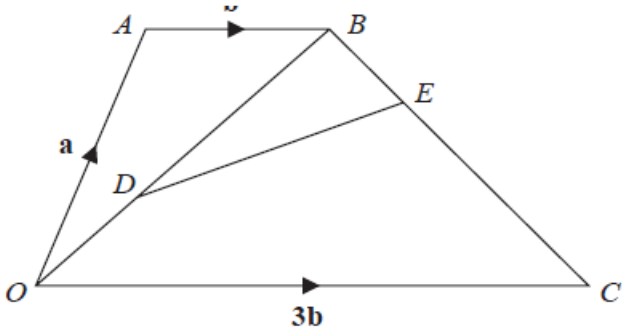
16 The curve **C** has equation $y = x^2 + 3x - 3$

The line **L** has equation $y - 5x + 4 = 0$

Show, algebraically, that **C** and **L** have exactly one point in common.

(Total for Question 16 is 4 marks)

18 *OABC* is a trapezium.



$\overrightarrow{OA} = \mathbf{a}$

$\overrightarrow{AB} = \mathbf{b}$

$\overrightarrow{OC} = 3\mathbf{b}$

D is the point on *OB* such that *OD* : *DB* = 2 : 3
E is the point on *BC* such that *BE* : *EC* = 1 : 4

Work out the vector $\overrightarrow{DE}$ in terms of **a** and **b**.
Give your answer in its simplest form.

.....

(Total for Question 18 is 4 marks)

20 Pat throws a fair coin n times.

Find an expression, in terms of n, for the probability that Pat gets at least 1 head and at least 1 tail.

.....

(Total for Question 20 is 2 marks)

22 Show that $\frac{6x^3}{(9x^2-144)} \div \frac{2x^4}{3(x-4)}$ can be written in the form $\frac{1}{x(x+r)}$ where r is an integer.

(Total for Question 22 is 3 marks)

TOTAL FOR PAPER IS 80 MARKS